The Mathematical Tools of Quantum Mechanics (Cohen–Tannoudji, Ch. II) Section C. Representations in the state space

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Outline

- A. One-particle wave function space
- B. State space and Dirac notation
- C. Representations in the state space
- D. Eigenvalue equations and observables
- E. Two key examples: $\{\ |\boldsymbol{r}\rangle\}\, and \{|\boldsymbol{p}\rangle\}; operators \boldsymbol{R},\ \boldsymbol{P}$
- F. Tensor products and applications

Definition of a representation

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- Vectors and operators are then represented in this basis by numbers: components for the vectors, matrix elements for the operators.
- We will reintroduce some concepts previously seen in Chapter A, using the Dirac notation: orthonormalization and closure relations, as well as the transformation from one representation to another.

Orthonormal sets and basis expansions

Orthonormalization relation

A set of kets, discrete $\{ |u_i\rangle \}$ or continuous $\{ |w_\alpha\rangle \}$, is said to be orthonormal if:

$$\boxed{\langle u_i | u_j \rangle = \delta_{ij}}$$

or

$$\boxed{\langle w_\alpha | w_{\alpha'} \rangle = \delta(\alpha - \alpha')}$$

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Highlight: For continuous sets, the $|w_\alpha\rangle$ are generalized kets.

Orthonormal sets and basis expansions

Basis property

A discrete set $\{|u_i\rangle\}$, or a continuous one $\{|w_\alpha\rangle\}$, constitutes a basis if every ket $|\psi\rangle$ has a unique expansion:

$$|\psi\rangle = \sum_i c_i \, |u_i\rangle$$

$$|\psi\rangle = \int d\alpha \; c(\alpha) \, |w_\alpha\rangle$$

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$$|\psi\rangle = \int d\alpha \; c(\alpha) \, |w_\alpha\rangle$$

$$\langle u_j|\psi\rangle = c_j \qquad \langle w_{\alpha'}|\psi\rangle = c(\alpha')$$

Closure relation and identity operator

Using the previous relations:

$$|\psi\rangle = \sum_i c_i \, |u_i\rangle = \sum_i \langle u_i|\psi\rangle \, |u_i\rangle = \sum_i |u_i\rangle\langle u_i|\psi\rangle = \left(\sum_i |u_i\rangle\langle u_i|\right) |\psi\rangle$$

$$\begin{aligned} |\psi\rangle &= \int d\alpha \, c(\alpha) \, |w_\alpha\rangle = \int d\alpha \, \langle w_\alpha|\psi\rangle \, |w_\alpha\rangle \\ &= \int d\alpha \, |w_\alpha\rangle\langle w_\alpha|\psi\rangle = \left(\int d\alpha \, |w_\alpha\rangle\langle w_\alpha|\right)|\psi\rangle \end{aligned}$$

Closure relation and identity operator

Closure (completeness) relations

$$\mathrm{P}_{(u_i)} = \sum_i |u_i\rangle\langle u_i| = \mathbb{1} \qquad P_{(w_\alpha)} = \int d\alpha\, |w_\alpha\rangle\langle w_\alpha| = \mathbb{1}$$

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Important: These relations express that $\{|u_i\rangle\}$ and $\{|w_\alpha\rangle\}$ constitute bases. Every ket has a unique expansion:

$$|\psi\rangle = \sum_i c_i |u_i\rangle \qquad |\psi\rangle = \int d\alpha\, c(\alpha)\, |w_\alpha\rangle \qquad \text{(C-14)}$$

Projectors and fundamental formulas

For $\mathbb{R}^3$ with orthonormal basis $\{e_1, e_2, e_3\}$:

$$P_1 + P_2 + P_3 = \mathbb{1}$$

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Fundamental formulas

$\{\ \lvert u_i\rangle\ \}$ representation	$\{\ \lvert w_\alpha\rangle\ \}$ representation
$\langle u_i \lvert u_j\rangle = \delta_{ij}$	$\langle w_\alpha \lvert w_{\alpha'}\rangle = \delta(\alpha - \alpha')$
$P_{(u_i)} = \sum_i \lvert u_i\rangle\langle u_i\rvert = \mathbb{1}$	$P_{(w_\alpha)} = \int d\alpha\ \lvert w_\alpha\rangle\langle w_\alpha\rvert = \mathbb{1}$

Representation of kets

In the $\{|u_i\rangle\}$ basis, a ket $|\psi\rangle$ is represented by the set of its components $c_i = \langle u_i|\psi\rangle$. These numbers can be arranged vertically to form a one-column matrix:

$$|\psi\rangle \longleftrightarrow \begin{pmatrix} \langle u_1|\psi\rangle \\ \langle u_2|\psi\rangle \\ \langle u_3|\psi\rangle \\ \vdots \end{pmatrix}$$

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In the continuous $\{|w_\alpha\rangle\}$ basis, the ket $|\psi\rangle$ is represented by a *continuous infinity* of numbers, $c(\alpha) = \langle w_\alpha|\psi\rangle$, that may be regarded as a function of α. We can imagine placing the values of $\langle w_\alpha|\psi\rangle$ along a vertical axis, producing a continuous "column" of components.

Representation of bras

Let $\langle\varphi|$ be an arbitrary bra. In the $\{|u_i\rangle\}$ basis,

$$\langle\varphi| = \langle\varphi|\mathbb{1} = \sum_i \langle\varphi|u_i\rangle\langle u_i|.$$

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The components of $\langle\varphi|$, namely $\langle\varphi|u_i\rangle$, are the complex conjugates of the components of the ket $|\varphi\rangle$: $b_i = \langle\varphi|u_i\rangle = \langle u_i|\varphi\rangle^*$.
In the same way, for the continuous basis $\{|w_\alpha\rangle\}$,

$$\langle\phi| = \int d\alpha\, \langle\phi|w_\alpha\rangle\langle w_\alpha|, \qquad b(\alpha) = \langle\phi|w_\alpha\rangle = \langle w_\alpha|\phi\rangle^*.$$

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We arrange the components of the bra horizontally to form a **row matrix**:

$$\begin{pmatrix}\langle\phi|u_1\rangle & \langle\phi|u_2\rangle & \langle\phi|u_3\rangle & \dots\end{pmatrix}$$

Bra–ket correspondence in matrix form

In the continuous case the components $\langle\phi|w_\alpha\rangle$ are placed along a horizontal axis, each value of α corresponding to a component of $\langle\phi|$:

$$(\ldots\ \langle\phi|w_\alpha\rangle\ \ldots)$$

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Comment: In a given representation, the matrices that represent a ket $|\psi\rangle$ and its associated bra $\langle\psi|$ are Hermitian conjugates of each other. Passing from one to the other corresponds to:

- interchanging rows and columns,
- taking the complex conjugate of each element.

Representation of operators by a "square" matrix

Given a linear operator A, in a discrete basis $\{|u_i\rangle\}$ we associate to it the set of numbers:

$$\boxed{A_{ij} = \langle u_i | A | u_j \rangle}$$

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These elements depend on two indices and can be arranged in a "square" matrix:

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

The first index (i) fixes the rows and the second (j) the columns.

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Each column j contains the components of the transform $A|u_j\rangle$.

Product of two operators

Using the closure relation, the matrix representing AB in the $\{|u_i\rangle\}$ basis is:

$$\langle u_i|AB|u_j\rangle = \langle u_i|\, A\mathbb{1}B\,|u_j\rangle = \sum_k \langle u_i|A|u_k\rangle\,\langle u_k|B|u_j\rangle = \sum_k A_{ik}B_{kj}.$$

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Key result

The matrix representing AB is the product of the matrices representing A and B:

$$(AB)_{ij} = \sum_k A_{ik}B_{kj}.$$

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Note: This convention is consistent with the usual matrix multiplication rule.

Matrix representation of the ket $|\psi'\rangle = A|\psi\rangle$

Given the components of $|\psi\rangle$ and of the operator A, the components c_i' of $|\psi'\rangle$ are:

$$c_i' = \langle u_i|\psi'\rangle = \langle u_i|A|\psi\rangle.$$

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Matrix form:

$$\begin{pmatrix} c_1' \\ c_2' \\ \vdots \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix}$$

Expression for the number $\langle\phi|A|\psi\rangle$

Inserting the closure relation between $\langle\phi|$ and A, and between A and $|\psi\rangle$:

$$\langle\phi|A|\psi\rangle = \sum_{i,j}\langle\phi|u_i\rangle\langle u_i|A|u_j\rangle\langle u_j|\psi\rangle = \sum_{i,j} b_i^* A_{ij} c_j,$$

where $b_i = \langle u_i|\phi\rangle$ and $c_j = \langle u_j|\psi\rangle$.

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Result: $\langle\phi|A|\psi\rangle$ is a number (a 1×1 matrix).

Outer product and operator construction

The expression $|\psi\rangle\langle\phi|$ is represented by a square matrix:

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Interpretation:

- The product $|\psi\rangle\langle\phi|$ is an operator.
- The product $\langle\phi|\psi\rangle$ is a number.

Matrix representation of the adjoint operator $A^\dagger$

From $\langle u_i|A^\dagger|u_j\rangle = \langle u_j|A|u_i\rangle^*$ we obtain:

$$(A^\dagger)_{ij} = A^*_{ji}.$$

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Therefore:

- The matrices representing A and $A^\dagger$ are Hermitian conjugates.
- One passes from one to the other by interchanging rows and columns and taking the complex conjugate.

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If A is Hermitian ($A^\dagger = A$):

$$A_{ij} = A^*_{ji}, \quad \text{and hence} \quad A_{ii} = A^*_{ii} \;\Rightarrow\; A_{ii} \in \mathbb{R}.$$

The diagonal elements of a Hermitian matrix are always real.

Change of representations: definition of the transformation matrix

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The change of basis is defined by the components:

$$S_{ik} = \langle t_i | u_k \rangle, \qquad S^{\dagger}_{ki} = \langle u_k | t_i \rangle.$$

S is the **transformation matrix**.

Unitarity of the transformation matrix

Using the closure relations

$$P_{(u)} = \sum_i |u_i\rangle\langle u_i| = \mathbb{1}, \qquad P_{(t)} = \sum_k |t_k\rangle\langle t_k| = \mathbb{1},$$

we obtain the orthonormality relations:

$$\langle u_i|u_j\rangle = \delta_{ij}, \qquad \langle t_k|t_l\rangle = \delta_{kl}.$$

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The transformation matrix S is unitary:

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Indeed,

$$(S^\dagger S)_{kl} = \sum_i S^\dagger_{ki} S_{il} = \sum_i \langle u_k|t_i\rangle\langle t_i|u_l\rangle = \langle u_k|u_l\rangle = \delta_{kl}.$$

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Similarly,

$$(SS^\dagger)_{ij} = \sum_k S_{ik} S^\dagger_{kj} = \sum_k \langle t_i | u_k \rangle \langle u_k | t_j \rangle = \langle t_i | t_j \rangle = \delta_{ij}.$$

Unitarity of the transformation matrix

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$$(SS^\dagger)_{ij} = \sum_k S_{ik} S^\dagger_{kj} = \sum_k \langle t_i | u_k \rangle \langle u_k | t_j \rangle = \langle t_i | t_j \rangle = \delta_{ij}.$$

Therefore, S preserves inner products: it is a unitary change of basis.

Transformation of the components of a ket

Let $|\psi\rangle$ be a ket with components $c_i = \langle u_i|\psi\rangle$ in the old basis. The components in the new basis $\{|t_k\rangle\}$ are:

$$\langle t_k|\psi\rangle = \sum_i \langle t_k|u_i\rangle\langle u_i|\psi\rangle = \sum_i S_{ki}c_i.$$

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Inverse transformation:

$$\langle u_i|\psi\rangle = \sum_k \langle u_i|t_k\rangle\langle t_k|\psi\rangle = \sum_k S^\dagger_{ik}\langle t_k|\psi\rangle.$$

Transformation of the components of a bra

The principle is analogous to that for kets. Let $\langle\phi|$ have components $b_i = \langle\phi|u_i\rangle$ in the old basis. In the new basis:

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Inverse transformation:

$$\langle\phi|u_i\rangle = \sum_k \langle\phi|t_k\rangle\langle t_k|u_i\rangle = \sum_k b'_k S_{ki}.$$

Transformation of the matrix elements of an operator

Let $A_{ij} = \langle u_i|A|u_j\rangle$ be the matrix elements of A in $\{|u_i\rangle\}$. In the new basis $\{|t_k\rangle\}$:

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Substitute the transformations:

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Substitute the transformations:

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Matrix form:

$$A' = SAS^{\dagger}.$$

(This is the standard rule for unitary change of basis.)